实测天体物理作业

1900011604 张植竣 北京大学

2022 年 3 月 21 日

1. 对于圆形孔径:

$$\mathrm{FWHP} = 58.4^\circ \times \frac{\lambda}{D} = 58.4^\circ \times \frac{c}{fD} = 1.75^\circ$$

2. 由源张角 $\theta_S = 30'$，主瓣张角 $\theta_B = 1.75^\circ = 105'$ 得主瓣温度:

$$T_B = T\left(\frac{\theta_S^2}{\theta_S^2 + \theta_B^2}\right) = 5800\,\mathrm{K} \times \left(\frac{30^2}{30^2 + 105^2}\right) = 437.7\,\mathrm{K}$$

天线温度:

$$T_A = \eta_B T_B = 0.7 \times 437.7\,\mathrm{K} = 306.4\,\mathrm{K}$$

3. 条纹间距

$$\delta = \frac{\lambda}{b} = \frac{c}{fb} = 61.9''$$

而复可见度函数:

$$V(u) = \int_{-\frac{\theta_S}{2}}^{\frac{\theta_S}{2}} AI_0 \mathrm{e}^{2\pi i u\theta}\,\mathrm{d}\theta, \quad u = \frac{b}{\lambda}$$

由于 A、I_0 均为常量，代入积分得:

$$V(u) = AI_0 \frac{\sin(\pi u \theta_S)}{\pi u} = -3.2 \times 10^5 AI_0$$